Housing Scarcity, Delayed First Births, and Fertility: Annualised Transition Extension

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Abstract

Birth rates across advanced economies have fallen in tandem with rising housing costs, yet standard macro-housing models treat household size as exogenous. We develop a life-cycle model in which politically motivated supply restrictions raise house prices, which delay first births through a crowding margin that reduces effective housing services, and delayed first births lower completed fertility because the reproductive window is finite. Persistent housing-scarcity shocks calibrated to a common 15 percent house-price target lower fertility consistently across three distinct supply-tightening margins—equivalent to roughly 0.05 TFR points, or about 13 percent of the US fertility decline since 2007—and the effect compounds over time as demographic aging and housing scarcity reinforce each other. In the steady-state comparison, higher house prices shift first births to older ages and more than double the childless share, with the fertility-aware political equilibrium settling at a lower house price than the benchmark without a fertility margin. The quantitative magnitudes are modest—housing scarcity alone does not explain the postwar fertility decline—but the channel is correctly signed, robust across supply margins, and represents a conservative lower bound because omitted channels (female labour supply, partner disagreement, delayed departure of adult children) each work in the same direction. An annualised transition extension, initialized at the observed age-by-tenure cross section, now also admits a stationary annual endpoint and a full rational-expectations transition around that endpoint. Within that annual closure, targeted help-to-buy support and lighter family deposit help operate mainly over a short housing-access crunch window: they raise young mortgaged ownership materially in the first five annual periods, with smaller but still positive effects at longer horizons. On the same upgraded annual permits/stock block, endogenous fertility inside the annual RE loop mainly shifts prices rather than ownership composition.

1 Introduction

Housing costs and fertility have moved in opposite directions across most advanced economies over the past half-century. Where housing is expensive and space is scarce, having a child becomes harder. Some births are delayed. Some are foregone entirely. A natural question is whether housing scarcity itself contributes to lower fertility, and if so, through what mechanism.

Answering that question requires a framework with two ingredients. First, the model needs a reason why housing remains scarce—not merely expensive, but persistently underbuilt because incumbent homeowners have a political incentive to restrict supply. Second, it needs a household problem in which children change the value of housing space. We build a life-cycle model with both features: endogenous land-use restrictions set through median-voter politics, competitive housing supply, and a household problem augmented with endogenous births, parity, children at home, and a crowding margin that lowers effective housing services when children are present. The supply side draws on the political-economy framework of Gross and Chivers (2025); the fertility block is new.

This is, to our knowledge, the first paper to embed endogenous fertility decisions—with explicit first-birth timing—into a political-economy housing model with endogenous land-use restrictions. The combination matters because it closes a demographic feedback loop: supply restrictions raise house prices, high prices delay first births, delayed births lower completed fertility, smaller families reshape the future electorate, and the changed electorate feeds back into housing policy. Existing work studies housing costs and fertility in reduced form (Clark, 2012; Dettling and Kearney, 2014) or embeds fertility in models without endogenous supply (Doepke, 2004). Our framework connects the two.

Three results organize the paper. First, persistent housing-scarcity shocks calibrated to a common 15 percent house-price target lower fertility robustly across three different supply-tightening margins. The quantitative effect is modest—a fertility reduction of roughly 0.007 at a 40-period horizon, equivalent to about 0.05 TFR points, or 13 percent of the US fertility decline since 2007—but consistent across margins and compounding over time. By 2100, the gap doubles as demographic aging and housing scarcity reinforce each other. This is the paper's main within-model counterfactual: the same economy under different supply regimes. Second, the model identifies a timing channel in the steady-state comparison: higher house prices shift first births to older ages, increase the share of first births after age 30, and lower the first-birth rate. When households account for the crowding cost of children, the political equilibrium settles at a lower house price than the benchmark without a fertility margin. Third, the available motivating evidence points in the same direction as the model. Across advanced economies, places with faster house-price growth also tended to experience larger delays in first births, and the historical aggregate comparison moves the same way. Those patterns are supportive but not decisive. The paper's contribution is therefore to identify and quantify a new channel from housing policy to demographic outcomes, not to rest on a fixed reduced-form design or to explain the full postwar fertility decline.

The paper proceeds as follows. Section 2 presents motivating facts on housing costs and fertility across advanced economies. Section 3 reviews the related literature. Section 4 sets out the model and its calibration. Section 5 reports the quantitative results, including steady-state first-birth timing, persistent housing scarcity, and the voter-preference comparison. Section 6 adds an annualised transition extension that starts from the observed age-by-tenure snapshot, anchors the annual branch on a stationary no-drift endpoint, and treats help-to-buy as a short-horizon coping mechanism rather than as the source of a new long-run annual benchmark. Section 7 concludes. The appendices collect the exact benchmark comparison, wide-range comparative statics, mechanism parameter sensitivity, and an endogenous departure envelope calculation.

2 Motivating Facts

Housing costs have risen sharply across advanced economies over the past three decades. At the same time, fertility has fallen below replacement in nearly all of them, and the age at which women have their first child has climbed steadily. This section documents three stylized facts that motivate the mechanism studied in the paper.

Fact 1: Real house prices have risen substantially, but heterogeneously

Figure 1 plots real house price indices for eight major economies, each re-indexed to $2005 = 100$. Several patterns stand out. First, the anglophone economies—the United States, the United Kingdom, Canada, and Australia—all experienced sustained real price appreciation, with Canada's index nearly tripling by the early 2020s. Second, trajectories varied widely. Japan's real house prices remained essentially flat over this period, reflecting the long aftermath of its 1990s asset-price

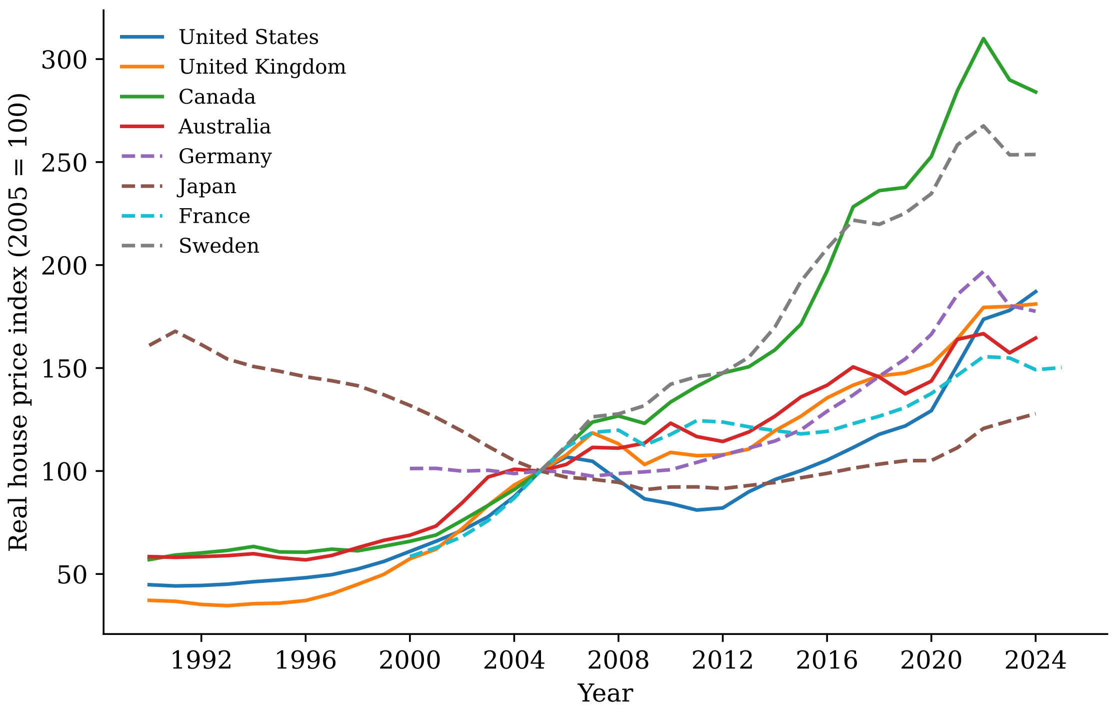

Figure 1: Real house price indices for eight major economies, indexed so that 2005 = 100. Sources: OECD Analytical House Price Indicators; FRED (Case-Shiller for US; BIS residential property prices for Japan, Canada, Australia).

collapse. Germany saw little real growth until approximately 2010, after which prices rose rapidly before correcting in 2022–2023. Sweden more than doubled in real terms. Third, despite this heterogeneity the overall direction is common: in seven of the eight countries shown, real house prices in the 2020s are materially above their mid-2000s level.

Fact 2: Fertility has fallen below replacement everywhere

Figure 2 shows total fertility rates (TFR) for the same eight countries from 1960 onward. The dashed line marks the conventional replacement level of 2.1 children per woman. By the 2020s every country in the sample is below replacement. Japan (TFR $\approx$ 1.20) and Canada ($\approx$ 1.26) are among the lowest. The United States, long an outlier among rich countries for its relatively high fertility, has fallen to roughly 1.6. Even countries that maintained near-replacement fertility into the 2000s—such as France, Australia, and Sweden—have declined noticeably in the past decade.

The timing of the decline differs across countries. Japan and Germany fell below replacement in the 1970s; the anglophone countries and France held near 2.0 until the late 2000s before declining. But the direction is universal. This common trend motivates a search for explanations that operate across institutional settings. Housing costs, present in every country to varying degrees, are a natural candidate.

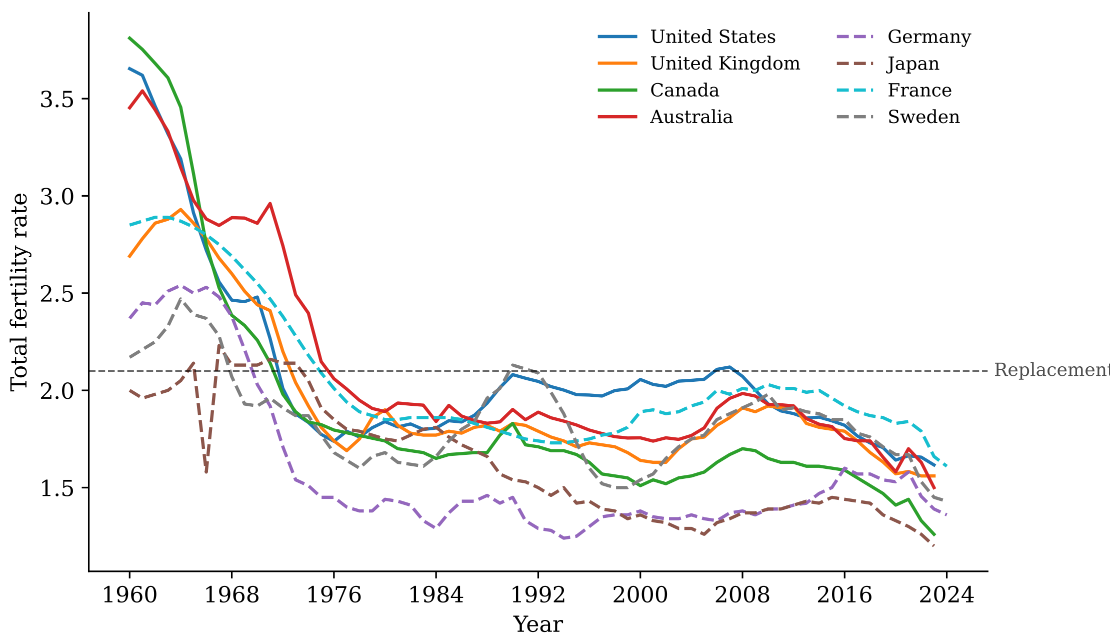

Figure 2: Total fertility rates for eight major economies, 1960–2024. The dashed line marks the conventional replacement level of 2.1. Sources: World Bank; Eurostat.

Fact 3: Countries with faster house price growth delayed first births more

Figure 3 links the preceding two facts. For each of the 27 European countries with at least five years of joint data on house prices and the mean age at first birth, we compute the cumulative real house price growth and the total change in the mean age at first birth over the country's observation window. The scatter reveals a positive association ($r = 0.50$): countries in which real house prices rose most also experienced the largest increases in the age at which women had their first child. Estonia, Lithuania, and Croatia—all with cumulative real price growth exceeding 150%—saw first-birth ages rise by more than 3.5 years. At the other extreme, Italy and the United Kingdom, with slower or more moderate price growth, recorded smaller shifts.

This cross-country association is, of course, only a correlation. It could be driven by income growth, urbanisation, educational expansion, or other omitted variables correlated with both housing costs and fertility timing. We do not interpret it as causal. Instead, we view it as a motivating pattern that a well-specified model should be able to rationalise. The structural model in Section 4 provides the link: when housing is scarce and expensive, young adults delay purchasing a home, which in turn delays—and in some cases discourages—the transition to parenthood.

These three facts—rising real house prices, falling fertility below replacement, and a positive cross-country association between housing cost growth and delayed first births—motivate a model in which housing scarcity operates as a binding constraint on family formation. The next section places these patterns in the related literature, and Section 4 then develops the model.

3 Related Literature

This paper contributes to several interconnected literatures spanning fertility, housing markets, and political economy. We organize the discussion around five strands and clarify how our framework

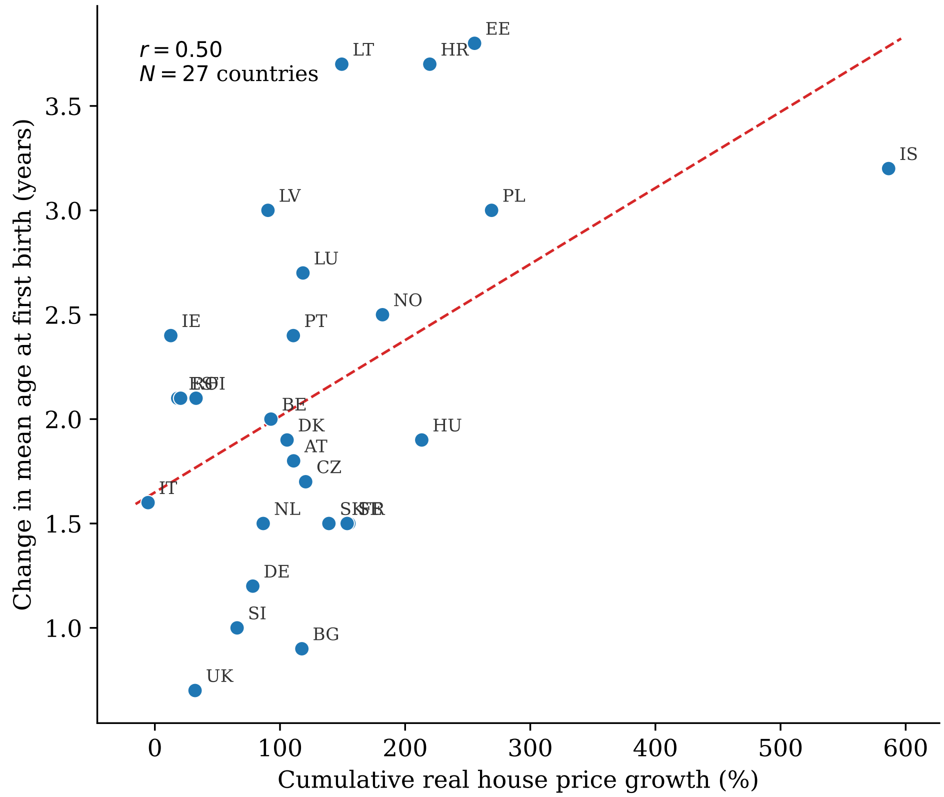

Figure 3: Cumulative real house price growth (%) vs. change in mean age at first birth (years) across 27 European countries. Turkey is excluded because its cumulative house-price growth exceeds 500%, roughly four times the next-highest country; including Turkey raises r to 0.54. Each point is one country; both variables are computed from the first to the last year in which house prices and first-birth age are jointly observed. The dashed line is a linear OLS fit; r is the Pearson correlation. Sources: OECD, Eurostat.

unifies them.

Fertility and housing costs. A growing body of evidence documents the link between housing costs and fertility outcomes. Ermisch (1988) provided early evidence that housing costs affect the timing of childbearing in the United Kingdom, showing that couples facing higher housing expenditure delay family formation. Simon and Tamura (2009) extended this line of inquiry to U.S. metropolitan areas, finding that local housing costs are negatively associated with fertility rates across cities. Dettling and Kearney (2014) sharpened the identification using MSA-level variation in house prices and established that rising house prices reduce birth rates among non-homeowners while modestly increasing them among owners, consistent with a wealth-effect channel operating alongside the cost-of-space channel. Clark (2012) documented similar patterns in the UK, emphasizing that binding housing constraints—particularly difficulty accessing owner-occupied housing—delay partnership formation and the transition to parenthood.

Lovenheim and Mumford (2013) exploited household-level variation in housing wealth from the PSID and found that a $100,000 increase in home equity raises the probability of having a child by 16–18 percent among homeowners, with no comparable effect for renters, confirming that tenure status mediates the fertility–house-price relationship. Daysal, Lovenheim, Siersbæk, and Wasser (2021) replicated this asymmetry in Danish register data and showed that housing wealth gains also improve infant health at birth, suggesting that the channel operates partly through resource availability rather than pure opportunity cost. These findings sit alongside a broader literature connecting local economic conditions to demographic outcomes. Autor, Dorn, and Hanson (2019) showed that adverse labor market shocks reduce marriage and fertility, highlighting how economic insecurity transmits into household formation decisions. Doepke and Kindermann (2019) developed a quantitative model in which disagreement between partners over the costs of childrearing contributes to fertility decline, underscoring that the decision to have children is shaped by both partners' perceptions of economic burden. Kearney, Levine, and Pardue (2022) surveyed the post-2007 U.S. birth-rate decline and concluded that no single economic or policy change accounts for the sustained fall; their analysis leaves open the possibility that persistent structural forces—such as housing scarcity—cumulate beneath the cohort-preference explanation.

Our paper builds directly on this strand by asking how a specific, persistent source of housing cost pressure—supply restrictions driven by political incumbents—affects not just the level of fertility but the timing of first births.

Birth timing and completed fertility. Our mechanism operates through delayed first births: when housing is expensive, households postpone the first child, and the finite reproductive window means some postponed births become foregone births. This builds on the quantity–quality framework of Becker (1960) and Becker and Lewis (1973), in which the shadow price of children rises with input costs. Hotz, Klerman, and Willis (1997) emphasized the role of tempo effects—delays in childbearing compress the remaining window for subsequent births and reduce completed family size. Goldin and Katz (2002) showed that access to the contraceptive pill enabled women to invest in careers, contributing to delayed family formation, while Caucutt, Guner, and Knowles (2002) showed that matching frictions lower fertility through a similar delay channel; both demonstrated that delay translates into lower completed fertility.

Several quantitative macro models have explored the aggregate implications of fertility choice. Greenwood, Seshadri, and Vandenbroucke (2005) explained the baby boom and bust through changes in the cost of household production, demonstrating that technology shocks that reduce the time cost of children can generate large swings in aggregate fertility—a mechanism that operates through the same shadow-price channel our model exploits, but on the technology side rather than the housing side. De la Croix and Doepke (2003) showed that differential fertility across income groups is a quantitatively important channel linking inequality to growth, establishing that models with endogenous fertility can generate macroeconomically significant effects even when individual-level responses appear small. Jones and Schoonbroodt (2016) embedded fertility in a dynastic model with aggregate shocks and showed that the Great Depression alone can generate a baby bust of 58 percent of the observed U.S. magnitude, followed by a substantial boom—confirming that fertility responds meaningfully to persistent economic conditions in calibrated general-equilibrium settings. Our contribution is to identify housing scarcity as an additional source of delay and to trace its aggregate fertility consequences in a political-economy framework where the source of scarcity is itself endogenous.

Housing supply regulation and NIMBYism. The source of housing scarcity in our framework is endogenous supply restrictions, a mechanism grounded in the housing supply regulation literature. Glaeser and Gyourko (2003) documented that land-use regulations raise housing costs substantially above construction costs in many U.S. cities. Glaeser, Gyourko, and Saks (2005) showed that the high cost of housing in Manhattan is attributable primarily to regulatory constraints rather than land scarcity or construction costs, and that these constraints have tightened over time. Saiz (2010) decomposed the housing supply elasticity into geographic and regulatory components, finding that regulation is the binding constraint in the most expensive markets. Gyourko, Hartley, and Krimmel (2021) updated and extended the Wharton Residential Land Use Regulatory Index to nearly 2,500 communities, documenting that the regulatory environment has tightened further since the mid-2000s and that the most regulated markets are disproportionately high-cost coastal cities. Turner, Haughwout, and van der Klaauw (2014) decomposed the welfare effects of land-use regulation into own-lot, external, and supply components, finding that the supply restriction effect generates large welfare losses—a result that motivates our focus on the quantity margin of housing supply rather than amenity externalities. The aggregate consequences of local supply restrictions are potentially large: Hsieh and Moretti (2019) estimated that housing constraints in high-productivity cities reduce U.S. aggregate output by misallocating labor across space.

Our paper takes the political-economy model of NIMBYism developed in Gross and Chivers (2025) as its upstream benchmark. That framework showed how majority-homeowner communities vote to restrict new construction in order to protect incumbents' house values, generating a wedge between efficient and equilibrium housing supply. We do not revisit that political-economy mechanism in detail here; rather, we embed it as the supply side of a richer demographic model and ask what happens to fertility when the housing stock is determined by incumbent voting.

Political economy of housing. The idea that homeowners have incentives to restrict supply has deep roots. Fischel (2001) articulated the "homevoter hypothesis," arguing that homeowners treat their house as their primary financial asset and vote on local land-use policy accordingly, favoring restrictions that protect property values. Ortalo-Magné and Prat (2014) formalized the political economy of urban growth, showing how the voting power of incumbent owners can generate inefficiently low levels of housing construction. Hankinson (2018) provided the first experimental evidence on NIMBYism, showing that even renters in high-cost cities oppose nearby construction at rates comparable to homeowners—evidence of scale-dependent preferences that help explain why supply restrictions persist even when a majority of residents support more housing in the abstract. Marble and Nall (2021) linked deed-level homeownership records to voter turnout for 18 million individuals and found that purchasing a home substantially increases participation in local elections, with the turnout boost nearly doubling when zoning issues are on the ballot, suggesting that homeowners' political engagement is partly asset-motivated. Parkhomenko (2023) embedded these mechanisms in a spatial equilibrium framework with heterogeneous agents, demonstrating that politically determined supply restrictions redistribute welfare from renters to owners and reduce aggregate housing consumption.

Our model inherits this political-economy structure but extends it to a setting where the demographic composition of the electorate—which depends on fertility outcomes—feeds back into housing policy. When supply restrictions delay childbearing, the resulting smaller cohorts alter the future balance of owners and renters, potentially reinforcing or attenuating the original supply constraint.

Demographic feedback into housing markets. Mankiw and Weil (1989) first argued that the baby-boom cohort's entry into the housing market drove the 1970s price boom, and that the subsequent bust would follow as smaller cohorts matured—a prediction that proved directionally correct even if its timing was off. Green and Hendershott (1996) refined the age–demand relationship, showing that the age profile of housing demand peaks in middle age and that shifts in cohort size generate persistent price effects. Chiuri and Jappelli (2003) documented that mortgage-market development shifts the age profile of homeownership toward the young across 14 OECD countries, implying that financial frictions interact with demographics to shape housing demand. Sommer and Sullivan (2018) showed that eliminating the U.S. mortgage interest deduction would lower homeownership rates and house prices substantially, providing a quantitative benchmark for the magnitude of policy-driven price effects. Mumtaz and Sustěk (2024) developed a model in which demographic structure is a first-order driver of house-price dynamics.

These papers treat demographics as exogenous. Our framework closes the loop: housing scarcity delays first births and reduces cohort size, which alters future housing demand and the political equilibrium that determines supply. The feedback is quantitatively modest in our calibration, but the mechanism is novel and correctly signed.

Contribution. To our knowledge, this is the first paper to embed an endogenous fertility decision—with explicit first-birth timing—into a political-economy model of housing supply with endogenous land-use restrictions. The fertility–housing-cost literature (strand one) has documented the empirical link between house prices and birth rates but has not modeled the supply-side origin of high prices. The birth-timing literature (strand two) has shown that delay reduces completed fertility but has not connected delay to a specific, policy-relevant source of housing cost pressure. The housing supply regulation and political-economy literatures (strands three and four) have modeled endogenous supply restrictions and their welfare consequences but have treated household size as fixed. The demographic feedback literature (strand five) has shown that cohort size drives housing demand but has taken demographics as exogenous. By connecting these strands, we can study a channel that none of them addresses in isolation: NIMBYism raises housing costs, which delays first births and reduces completed fertility, which reshapes the future electorate and housing demand. The quantitative framework allows us to assess the magnitude of this channel and to evaluate policies that target the supply side of the housing market as instruments of demographic policy.

4 Model

This section presents the full model. The framework nests the political-economy housing-supply model of Gross and Chivers (2025) and augments the household problem with endogenous fertility, children at home, and a crowding margin that reduces effective housing services. When the fertility block is shut off, the model reproduces the earlier benchmark exactly. When it is active, the household problem changes in two ways: family state enters the value function, and a new birth decision generates endogenous fertility outcomes across the life cycle.

4.1 Environment

The economy is populated by overlapping generations of households who live from age 25 to age 90 in five-year periods, giving 14 periods of active life. In each period a household chooses non-housing consumption, housing tenure (rent or own), net financial wealth, and whether to have a child. Housing supply is determined by competitive real-estate firms that operate subject to political constraints. A median-voter condition applied to the stationary distribution of household preferences determines the equilibrium level of supply restrictions and, through them, the equilibrium house price.

The environment has three blocks: (i) a household problem with endogenous fertility, (ii) a competitive real-estate sector, and (iii) a political equilibrium that closes the model. The household problem is the main object of interest. The real-estate and political blocks follow Gross and Chivers (2025) without modification.

4.2 Household problem

A household entering a period is characterized by the state vector (b, a, z, h, p, t), where b is net financial wealth, a is housing services, z is an idiosyncratic income state following an age-dependent Markov chain, $h \in \{0, 1, 2, 3\}$ is the number of children currently at home, $p \in \{0, 1, 2, 3^+\}$ is parity (children ever born), and t is the household's age. The distinction between h and p matters for the mechanism. Parity is permanent: once a child is born, p increases for the rest of the household's life. Children at home h is temporary: a child raises housing demand only while living at home and can leave according to a reduced-form departure schedule.

Household income is $y_t = e^{z_t} \cdot \zeta_t$, where ζ_t is a deterministic life-cycle profile. The income process is inherited from Gross and Chivers (2025). The household can rent or own housing, subject to a loan-to-value constraint ($-b \leq q \cdot a \cdot 0.90$) and a two-percentage-point borrowing spread. The housing tenure choice—rent, hold, or adjust—follows the same structure as Gross and Chivers (2025), including the rental services discount ($\theta_r = 0.85$) and the housing adjustment cost ($\kappa_a = 0.06$). The full budget constraints and rate parameters are reported in Table 1.

4.3 Effective housing services and crowding

The central innovation of the fertility extension is that children at home reduce effective housing services. The household values effective housing

$$H_t^{\text{eff}} = \frac{H_t}{(1 + \lambda_c \, h_t)^{\psi_c}}, \tag{1}$$

where H_t is raw housing services, h_t is the number of children currently at home, $\lambda_c = 0.18$ governs the strength of the crowding effect, and $\psi_c = 0.70$ controls its curvature.

This specification generates two channels through which children affect housing demand:

1. *Direct crowding.* A child at home reduces effective space today. For a household with one child at home, effective housing falls by a factor of $(1 + 0.18)^{0.70} \approx 1.125$, so the household loses roughly 12.5 percent of its effective housing services. A second child at home reduces effective housing by a factor of $(1 + 0.36)^{0.70} \approx 1.243$, a further decline. The direct channel raises the shadow cost of housing space whenever children are present.

2. *Anticipatory delay.* The household is forward-looking and understands that a birth today will crowd the dwelling in future periods. A household that anticipates wanting multiple children

has an incentive to delay the first birth until it can afford more housing space, because the crowding penalty applies to all children simultaneously at home. This anticipatory channel generates the first-birth delay that is the paper's main object of interest.

The two channels interact. The direct channel means that children make housing more expensive in the period they are at home. The anticipatory channel means that the expectation of this cost feeds backward into the birth decision, causing households to postpone fertility until they occupy larger or cheaper housing.

4.4 Flow utility

In a period without a birth, the household receives flow utility

$$u_t^{F,0} = \log c_t + s_h \log H_t^{\mathrm{eff}} + \nu_h\, h_t - \mathrm{penalty}_t, \tag{2}$$

where $s_h = 1$ is the weight on housing services in utility, $\nu_h = 0.02$ is the direct flow value of having children at home, and $\mathrm{penalty}_t$ is a large penalty imposed when the loan-to-value constraint is violated.

When a birth occurs, the newborn enters both the crowding state and the family-utility term immediately:

$$u_t^{F,1} = \log c_t + s_h \log\!\left(\frac{H_t}{(1+\lambda_c(h_t+1))^{\psi_c}}\right) + \nu_h(h_t+1) + \phi(p_t) - \kappa_0 - \kappa_q\, q_t - \mathrm{penalty}_t. \tag{3}$$

The birth-period utility differs from the no-birth utility in four terms:

- The crowding denominator uses $h_t + 1$ rather than h_t, reflecting the newborn's immediate presence in the dwelling.
- The child-at-home flow value applies to $h_t + 1$ children.
- The household receives parity-specific birth utility $\phi(p_t)$, which captures the value of having the $(p_t + 1)$-th child. The calibrated values are $\phi(0) = 1.05$ for the first birth, $\phi(1) = 1.15$ for the second, and $\phi(2) = 0.95$ for the third.
- The household pays a common birth cost $\kappa_0 = 0.06$ and a price-sensitive birth cost $\kappa_q\, q_t$ with $\kappa_q = 0.24$. The price-sensitive term $\kappa_q q_t$ is a reduced form for the resource cost of space reallocation at birth—the cost of moving to a larger dwelling, furnishing a child's room, or absorbing higher rent on the marginal square metre. While the linear specification is a simplification, the implied semi-elasticity is consistent with external evidence: Dettling and Kearney (2014) estimate that a 10 percent increase in local house prices reduces birth rates among renters by approximately 3–5 percent, which corresponds to the range of κ_q values explored in Appendix C.

The birth decision is modeled as a logistic choice. Conditional on the household's state, the probability of a birth is

$$\Pr(\mathrm{birth}) = \frac{1}{1 + \exp\!\big(-(V^{\mathrm{birth}} - V^{\mathrm{no\ birth}})/\sigma\big)}, \tag{4}$$

where V^{birth} and $V^{\mathrm{no\ birth}}$ are the continuation values associated with having and not having a birth, and $\sigma = 0.15$ is a scale parameter that governs the smoothness of the birth choice. The logistic specification avoids the bang-bang behavior of a discrete choice and generates interior birth probabilities that vary continuously with the household's state.

4.5 Child dynamics

Children leave home according to a reduced-form departure schedule calibrated to US Census co-residence rates. The schedule is defined over three departure ages, each measured from the child's birth:

Departure age (child's age)	20	25	30
Departure probability	0.588	0.147	0.265

These probabilities imply that a child born when the parent is age 25 may leave home when the parent is 45, 50, or 55. Because the model uses five-year periods, the departure schedule generates a well-defined transition for h_t: in each period, the number of children at home can decrease by one as earlier children depart, while it can increase by one if a birth occurs.

Parity p never decreases. It records the permanent count of children ever born. Children at home h can decrease as children leave. This asymmetry is what allows the model to distinguish between completed fertility (a parity outcome) and contemporaneous crowding (a children-at-home outcome).

Births can occur at parent ages 25, 30, 35, and 40. The finite reproductive window is a biological constraint: fecundity declines sharply after age 35 and is near zero by age 45 (Leridon, 2004; te Velde and Pearson, 2002). In the current benchmark, parity-zero births also use age-specific realized-birth shifters calibrated to the recent US first-birth distribution, so the entry-into-parenthood hazard is no longer flat within the birth window. This means that the birth window spans four of the fourteen life-cycle periods, but later first births are less likely to realize than earlier first births. After age 40, no further births are possible, but children may still be at home and continue to affect housing demand through the crowding channel.

4.6 Bequest and terminal condition

At the terminal age (90), the household receives bequest utility

$$u^{\text{bequest}} = \beta_{\text{beq}} \cdot \log(\text{net worth}), \tag{5}$$

where $\beta_{\text{beq}} = 0.98$ and net worth is the sum of financial wealth and housing equity. The value function is solved by standard backward induction from the terminal period. In each period, the household evaluates each feasible combination of housing tenure, housing level, bond holdings, and birth decision, and selects the combination that maximizes expected discounted utility.

4.7 Real estate firms

The supply side follows Gross and Chivers (2025) without modification. Competitive real-estate firms supply housing. Each firm takes the house price q as given and chooses how much to build. The aggregate supply function depends on the house price and on the regulatory environment, which is determined endogenously through the political block. Higher house prices induce more construction, but the response is attenuated by political restrictions on new supply.

4.8 Housing supply and median voter

The political equilibrium determines the level of supply restrictions. Following Gross and Chivers (2025), each household evaluates the effect of a small persistent increase in the house price on its expected lifetime utility. Specifically, the model computes the value function at the current price q and at a perturbed price $q \cdot d$, where $d = 1.01$ represents a one-percent persistent price increase. A household supports tighter supply restrictions—and therefore higher house prices—if $V(q \cdot d) > V(q)$.

Aggregating these binary preferences across the stationary distribution of households yields a vote object. The median-voter condition requires that the aggregate vote switches sign at the equilibrium price: below the crossing price, a majority of households (weighted by stationary mass) support tighter supply; above it, a majority opposes further restrictions.

With the fertility extension, family state changes voter preferences. A young renter planning a first birth values lower house prices more than the same renter without children would, because the renter internalizes both the direct housing cost and the crowding penalty that a child will impose. Family-forming households are less willing to tolerate high house prices once they account for both housing costs and the effect of crowding on effective space. Section 5.3 quantifies the resulting shift in the political crossing price.

4.9 Equilibrium definition

A stationary equilibrium consists of:

1. A house price q and a rental rate r.

2. Value functions $V(b, a, z, h, p, t)$ for each age t and family state (h, p), solving the household's dynamic programming problem by backward induction.

3. Policy functions for consumption, housing tenure, housing level, bond holdings, and birth probability, all functions of the state (b, a, z, h, p, t).

4. A stationary distribution $\mu(b, a, z, h, p, t)$ over household states, generated by forward iteration of the policy functions and child-departure transitions from an initial distribution of young households.

5. A median-voter condition: the aggregate vote object

$$\mathcal{V}(q) = \sum_{b,a,z,h,p,t} \mu(b, a, z, h, p, t) \cdot \operatorname{sign}\big[V(b, a, z, h, p, t; q \cdot d) - V(b, a, z, h, p, t; q)\big] \tag{6}$$

   switches sign at the equilibrium price q^*. The crossing price q^* is where $\mathcal{V}$ passes through zero.

In the NIMBY benchmark without fertility, the state space is (b, a, z, t) and the vote object aggregates over that smaller distribution. The fertility extension enlarges the state space and changes the value function, but the equilibrium definition retains the same structure.

4.10 Calibration

The model inherits all housing, income, voting, and supply parameters from the NIMBY benchmark of Gross and Chivers (2025). Table 1 reports these inherited parameters. The fertility extension adds a second set of parameters governing births, child dynamics, and crowding. Table 2 reports those parameters.

The key calibration targets for the fertility block are the parity distribution at age 50, targeted to approximate US Census shares. Table 3 reports the target and model moments. The model matches the broad shape of the parity distribution—the two-child mode and the declining tail—with the main discrepancy being an overprediction of childlessness (0.312 versus the 0.165 target) and a

Table 1: Parameters inherited from the NIMBY benchmark (Gross and Chivers, 2025)

Parameter	Value	Description
Discount factor β	0.98	Standard life-cycle calibration
Housing preference s_h	1	Weight on housing in utility
Borrowing spread r_{spread}	0.02	Wedge between borrowing and saving rates
Risk-free rate r_a	-0.03	Net of depreciation
Rent markup	0.02	Rental rate markup over user cost
LTV ratio	0.90	Maximum loan-to-value
Adjustment cost κ_a	0.06	Housing transaction cost
Bequest weight β_{beq}	0.98	Terminal bequest utility weight
Age range	25–90	In 5-year periods (14 periods)
Financial grid	60 points	$b \in [-20, 20]$
Housing grid	14 points	$a \in [0, 15]$
Rental option	$\theta_r = 0.85$	Rental housing services fraction

Table 2: Fertility-extension calibration

Parameter	Value	Role
Birth utility $\phi(p)$ by parity	1.05, 1.15, 0.95	Parity-specific birth incentives
Child-at-home utility ν_h	0.02	Direct flow value of children while at home
Common birth cost κ_0	0.06	Baseline utility cost of a birth
Price-sensitive birth cost κ_q	0.24	Links house prices to fertility decisions
Crowding parameter λ_c	0.18	How children reduce effective housing
Crowding curvature ψ_c	0.70	Curvature of crowding function
Logistic scale σ	0.15	Smoothness of birth choice
Leave-home ages	20, 25, 30	Reduced-form departure grid
Leave-home probabilities	0.588, 0.147, 0.265	Calibrated to Census co-residence rates
Birth-eligible ages	25, 30, 35, 40	Age-specific realized-birth shifters for parity zero: $[1.00, 0.55, 0.22, 0.06]$; unit weights for higher parities
Max children at home C	4	Including 0 (so 0–3 children)
Max parity P	4	Including 0 (so 0–3+ children)

corresponding underprediction of three-or-more. The two most important mechanism parameters are the price-sensitive birth cost κ_q and the crowding parameter λ_c. The price-sensitive birth cost governs the direct channel from house prices to fertility decisions: a higher κ_q makes births more costly when housing is expensive. The crowding parameter governs the indirect channel through which children reduce effective housing services: a higher λ_c strengthens the crowding penalty and amplifies the incentive to delay births until the household can afford more space. Together, these two parameters determine the quantitative strength of the fertility–housing mechanism.

Table 3: Calibration fit: parity distribution at age 50

	Childless	1 child	2 children	3+ children
US Census target	0.165	0.193	0.357	0.285
Model (at $q \approx 1.824$)	0.312	0.079	0.449	0.160

4.11 Nesting result

A methodological property of the fertility extension is that it nests the NIMBY benchmark of Gross and Chivers (2025) exactly. When the fertility block is shut off—setting $C = 1$, $\phi(p) = 0$, $\nu_h = 0$, $\kappa_0 = 0$, $\kappa_q = 0$, $\lambda_c = 0$, and $\psi_c = 0$—the fertility solver reproduces every upstream object: the distance, the aggregate vote, and the debt stock all match at the same (q, r_b^+) to numerical precision. This exact nesting is not automatic in models that augment the state space. It requires the fertility extension to enter the household problem additively and to leave the real-estate and political blocks unchanged. The payoff is a clean identification claim: any difference in outcomes between the fertility model and the NIMBY benchmark is driven entirely by the fertility channel, not by a different equilibrium definition, supply structure, or numerical approximation.

5 Quantitative Results

This section presents the model's quantitative predictions. Section 5.1 examines steady-state first-birth timing. Section 5.2 turns to persistent housing scarcity. Section 5.3 compares voter preferences in the fertility and NIMBY benchmarks. Section 5.4 closes with brief supporting evidence.

5.1 Steady-state first-birth timing

The cleanest result from the fertility extension is a steady-state comparison across house-price levels. Because parity is an explicit household state, the model can trace first births directly rather than only tracking total or completed fertility. Figure 4 plots the three timing objects—mean age at first birth, share of first births at age 30 or later, and the first-birth rate—against the steady-state house price q.

The central result is a comparison of fertility outcomes at the two political-equilibrium prices. The NIMBY benchmark (fertility shut off) crosses at $q = 2.326$; the refreshed fertility benchmark now crosses locally at $q = 1.750$. Table 4 reports the demographic consequences of this gap, evaluating the fertility model at those two benchmark prices. Moving from the fertility-model crossing to the higher NIMBY crossing price delays mean first birth by about 1.4 years, doubles the share of first births at age 30 or later, raises the childless share by about 19.8 percentage points, and lowers the first-birth rate from about 0.213 to about 0.163. The large completed- fertility adjustment comes mainly through a collapse in the share of households with three or more children, from about 41.7 to about 15.3 percent. These are still meaningful differences in completed family size, but the cleaner and better-disciplined model object is the timing margin itself: expensive housing pushes entry into parenthood later, and that delay carries through into fewer realized births by age 50.

Table 4: Fertility-model outcomes at the two benchmark crossing prices

	Fertility crossing ($q = 1.750$)	NIMBY crossing ($q = 2.326$)	Difference
First-birth timing			
Mean age at first birth	26.22	27.60	+1.38 years
Share of first births at 30+	0.202	0.405	+0.204
First-birth rate	0.213	0.163	−0.050
Parity distribution at age 50			
Childless	0.148	0.347	+19.8 pp
One child	0.027	0.075	+4.7 pp
Two children	0.407	0.425	+1.8 pp
Three or more	0.417	0.153	−26.4 pp

This comparison characterises how the political equilibrium shifts when the household problem includes fertility, not a policy counterfactual. The NIMBY benchmark has no fertility margin—it is a different model, not a world where housing policy ignores fertility. The within-model counterfactual is the scarcity experiment in Section 5.2, which compares the same fertility-aware economy under different supply regimes.

The mechanism operates through both the direct and indirect housing channels identified in the model section. The price-sensitive birth cost $\kappa_q q_t$ makes births more expensive in high-price environments. The crowding margin λ_c raises the shadow cost of space when children are present. Together, these channels push first births later and reduce the share of households who have children at all.

Figure 4 plots the three timing objects against the full range of steady-state house prices. The gradients are steep: moving from $q = 1.50$ to $q = 3.00$ raises mean first-birth age by about 4.0 years and cuts the first-birth rate by more than half. The wide-range comparative statics are reported in Appendix Table 10.

5.2 Persistent housing scarcity and fertility

A single political-tightness parameter is not enough for the claim that housing scarcity lowers fertility, because it embeds the supply-tightening result in one dial that may be doing too much work. A more disciplined comparison calibrates three different supply-tightening shocks to the same housing outcome: a 15 percent higher house-price index after 40 periods. The three margins are:

1. A reduction in supply elasticity (ϵ_0 reduced by 6.2 percent);
2. A reduction in baseline supply (s_0 reduced by 2.4 percent);
3. A shift in baseline political tightness (θ_0 increased by +0.019).

This calibration strategy does two things. First, it shows that the result is not tied to a single primitive. Second, it anchors the long-run exercise to a common housing target rather than to an arbitrary parameter movement. Figure 5 reports the resulting comparison.

Table 5 summarizes the quantitative outcomes. At horizon $t = 40$, all three shocks lower the fertility rate from 0.248 to 0.241—a reduction of roughly 0.007. The effect is modest but remarkably

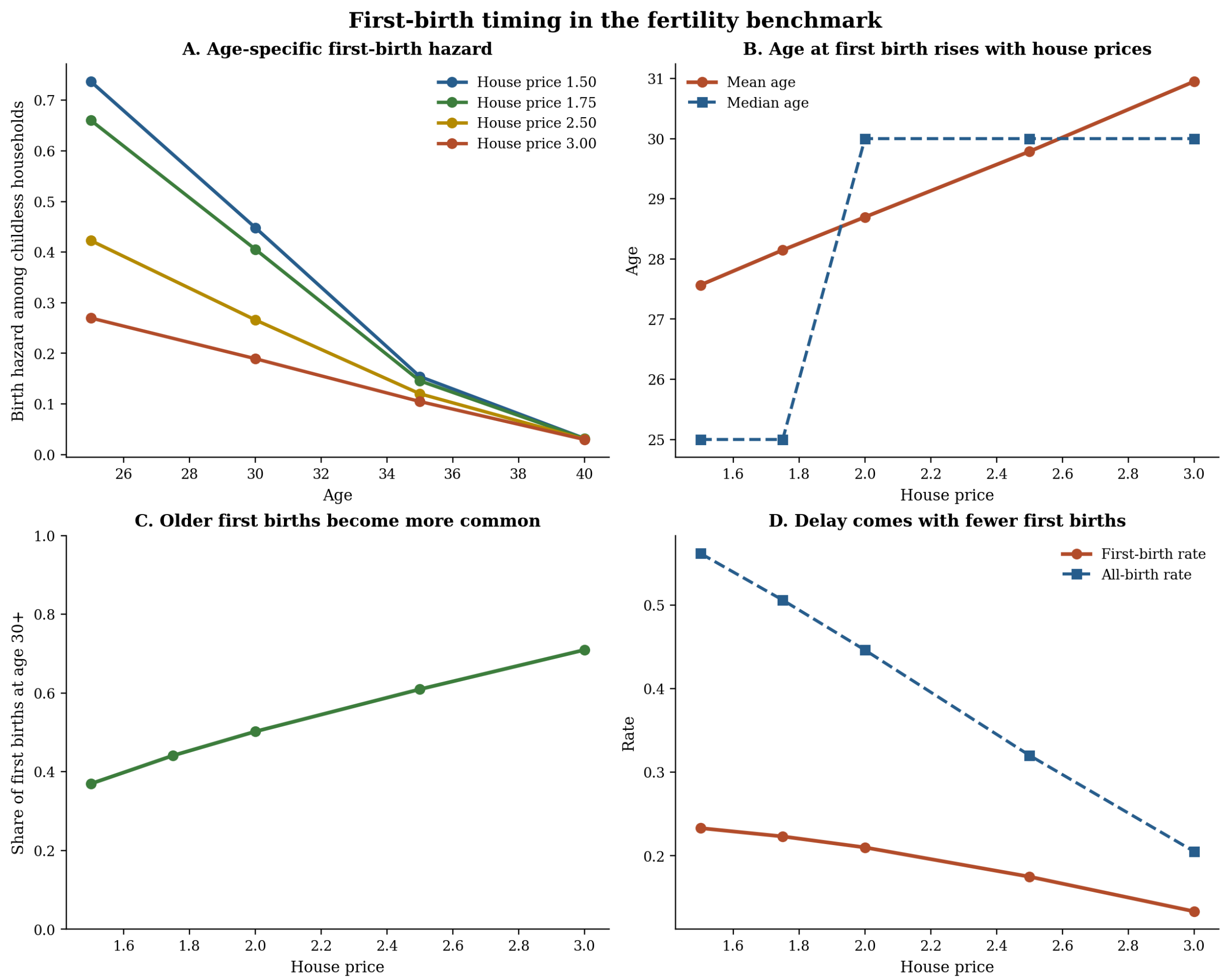

Figure 4: First-birth timing in the fertility benchmark. Higher house prices shift first births to older ages, increase the share of first births occurring at age 30+, and lower the first-birth rate.

consistent across margins. On the historical comparison, the calibrated shocks improve RMSE against the aggregate 1956–1995 fertility decline by 3.4 to 4.0 percent and close roughly 5.4 percent of the baseline-to-data end-gap.

Table 5: Persistent housing scarcity: three supply-tightening margins

Supply margin	Calibration	Fertility at $t = 40$		RMSE
		Baseline	Shock	improvement
ϵ_0 reduction	ϵ_0 reduced 6.2%	0.248	0.241	4.0%
s_0 reduction	s_0 reduced 2.4%	0.248	0.241	3.6%
θ_0 shift	θ_0 +0.019	0.248	0.241	3.4%

To put the 0.007 reduction in perspective: the model birth rate of 0.248 corresponds roughly to a total fertility rate of 1.7 children per woman.[1] A 2.8 percent reduction therefore translates to approximately 0.05 TFR points. Applied to the approximately 3.6 million annual US births, this implies roughly 100,000 fewer births per year attributable to the housing-scarcity channel alone. Against the recent US fertility decline (TFR fell from approximately 2.0 in 2007 to 1.62 in 2023, a drop of 0.38 points), the housing channel accounts for roughly 13 percent of the observed decline—modest but not negligible for a single mechanism operating through one margin of housing policy.

The effect also compounds over time. Table 6 reports the forward projections under a medium-immigration demographic scenario. At the 40-period calibration horizon, the fertility gap is 0.007 across all three margins. By 2050, the gap narrows slightly as the direct shock effect stabilizes. But by 2100, the gap widens to 0.006–0.014, reflecting the interaction between housing scarcity and demographic aging: restricted supply raises house prices further, which lowers realized fertility below what aging alone would produce.

Table 6: Compounding effect: fertility gap widens over time

	Baseline	ϵ_0 shock	s_0 shock	θ_0 shock
Fertility rate				
$t = 40$ (calibration)	0.248	0.241	0.241	0.241
2050 projection	0.194	0.191	0.190	0.192
2100 projection	0.188	0.179	0.174	0.182
House-price index				
$t = 40$ (calibration)	1.000	1.150	1.150	1.150
2050 projection	1.066	1.134	1.162	1.103
2100 projection	1.254	1.606	1.819	1.450

These projections are illustrative; at a 75-year horizon, structural changes in technology, preferences, and policy make point forecasts unreliable.

The widening price gap in the bottom panel of Table 6 explains the compounding: each supply margin that starts at a 15 percent price premium amplifies over time, and the s_0 reduction reaches an 82 percent premium by 2100.

[1] The transition birth rate of 0.248 is a population-weighted average across all age groups, including those past childbearing. With the model's demographic structure, this maps to approximately 1.7 births per woman over the reproductive window. The parity distribution at the steady-state crossing implies completed fertility of approximately 1.46, which is lower because the crossing is evaluated at a single price level rather than along the transition path.

Housing scarcity therefore moves fertility in the right direction, and the effect grows rather than dissipates. The model is better understood as identifying a compounding channel—persistent underbuilding delays and discourages family formation, and the demographic consequences feed back into housing demand—rather than as a complete account of postwar fertility trends.

This also clarifies the relationship to the baby-boom exercise in Gross and Chivers (2025). The baby-boom experiment asks how a larger cohort pushes on housing demand, prices, and politics over time. The present paper asks the reverse question: how tighter housing supply pushes back on fertility. The two mechanisms are therefore complementary, not competing. A large cohort raises housing pressure through the children-at-home demand channel; sustained housing scarcity then feeds back into later and lower realized fertility. In that sense, the baby-boom transition is best read as a dynamic stress test of the family-housing-demand mechanism, while the persistent-scarcity exercise is the main causal direction emphasized here.

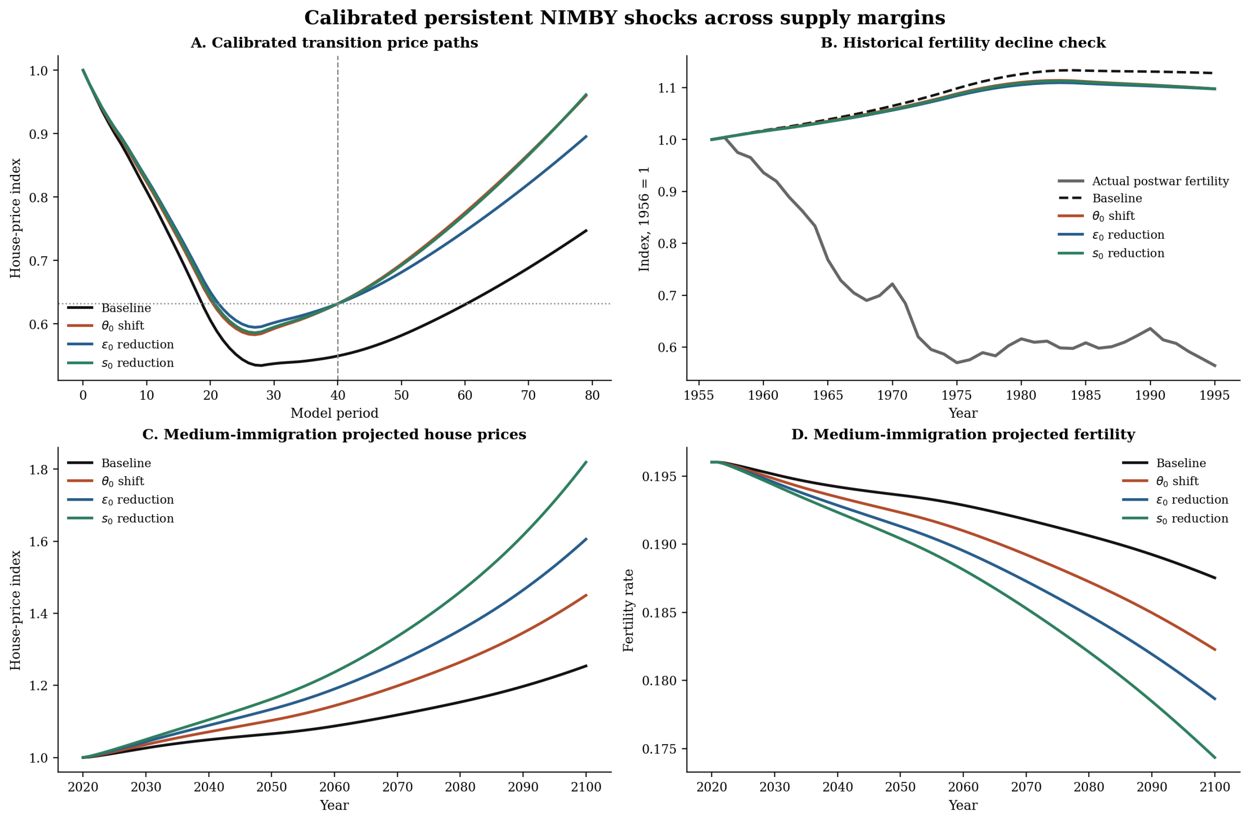

Figure 5: Persistent housing scarcity calibrated to a common housing target. Three different supply-tightening shocks are chosen to raise the 40-period house-price index by 15 percent. All three lower fertility.

5.3 Voter preference comparison

The fertility extension changes the political equilibrium because family-forming households evaluate house-price increases differently from otherwise identical households without children. Table 7 reports the net support index at selected price levels in both the NIMBY benchmark (fertility shut off) and the fertility benchmark (fertility active). The reported object is vote per unit mass, which ranges from -1 (all households oppose higher prices) to $+1$ (all households support them). A positive value means a net majority supports tighter supply restrictions at that price.

Table 7: Net support for higher prices: NIMBY vs. fertility benchmarks

House price q	NIMBY	Fertility	Interpretation
1.50	+0.84	+0.22	Both support higher prices; fertility support is much weaker
1.75	+0.79	−0.00	NIMBY still supports; fertility is essentially at its local crossing
2.25	+0.23	−0.28	NIMBY still supports; fertility opposes
2.50	−0.53	−0.75	Both oppose; fertility more strongly

The comparison reveals a sharp divergence, but the headline should come from the local benchmark search rather than from a monotone reading of the coarse common grid. On the common grid, households in the fertility benchmark are already close to indifferent at $q = 1.75$ and oppose further price increases by $q = 2.25$, while the NIMBY benchmark still supports higher prices at that point. The local search then places the political crossing—the price at which the median voter switches from supporting to opposing tighter supply—at 2.326 in the NIMBY benchmark versus 1.750 in the fertility benchmark.

The mechanism is straightforward. When household utility depends on family state, younger households planning children place a higher value on affordable housing. The crowding margin amplifies this: each additional child at home raises the shadow cost of housing space, making price increases more costly. The fertility benchmark therefore shifts political weight toward lower prices.

Two caveats apply. First, this is a within-model comparison, not an empirical claim about actual voting behavior. The model does not include an explicit ballot or referendum mechanism; it uses the same median-voter condition as Gross and Chivers (2025). Second, the comparison assumes the same income process and demographic structure in both benchmarks. In practice, the fertility decision itself could feed back into income and wealth accumulation in ways the current model does not capture. The comparison is best read as showing that endogenous fertility shifts the political equilibrium toward more permissive supply, not as a prediction about the magnitude of that shift in any specific jurisdiction.

5.4 Supporting evidence

The model implies that expensive housing should delay first births and lower fertility. The paper does not rest on a settled empirical design, but two supporting pieces of evidence point in the same direction as the model. The goal here is discipline and sign-checking, not a definitive causal estimate.

Two pieces of evidence point in the same direction as the model. First, a panel of 28 European countries with joint OECD house-price and Eurostat first-birth data (486 country-year observations, country and year fixed effects) shows that higher real house prices predict later first births: a one-unit increase in the OECD real house-price index is associated with a 0.00033-year increase in the mean age at first birth ($p = 0.038$). In a parallel specification covering 557 country-year observations, the same index predicts lower total fertility ($\hat{\beta} = -0.00022$, $p = 0.006$). Both regressions include country and year fixed effects, which absorb most of the overall variation. The implied magnitudes are small—a one-standard-deviation increase in the house-price index shifts mean first-birth age by roughly 0.03 years—but the signs are correct and statistically significant. This is best read as supportive evidence for the mechanism rather than as a tight calibration target: within-country, within-year variation in house prices predicts the timing and level of fertility in the direction the model implies, but the design is not the paper's main contribution.

Second, the aggregate postwar comparison provides limited but informative context. Three calibrated housing-scarcity shocks—each designed to raise the house-price index by 15 percent—improve the model's fit to the 1956–1995 fertility decline by 3.4 to 4.0 percent RMSE and close roughly 5.4 percent of the baseline-to-data end-gap. Figure 6 shows that the fertility extension fits the aligned post-boom decline modestly better than the flat return path of the benchmark model. Housing scarcity alone does not explain the full decline, but it adds a margin in the right direction.

The overall evidence is consistent with the model but not yet conclusive. The cross-country panel confirms the sign of the housing–fertility relationship but does not identify the mechanism cleanly, since house prices covary with income, urbanisation, and educational attainment. A fuller empirical design is left for future work. For the current paper, these facts are best read as supporting context around a model-driven contribution rather than as a fixed empirical core.

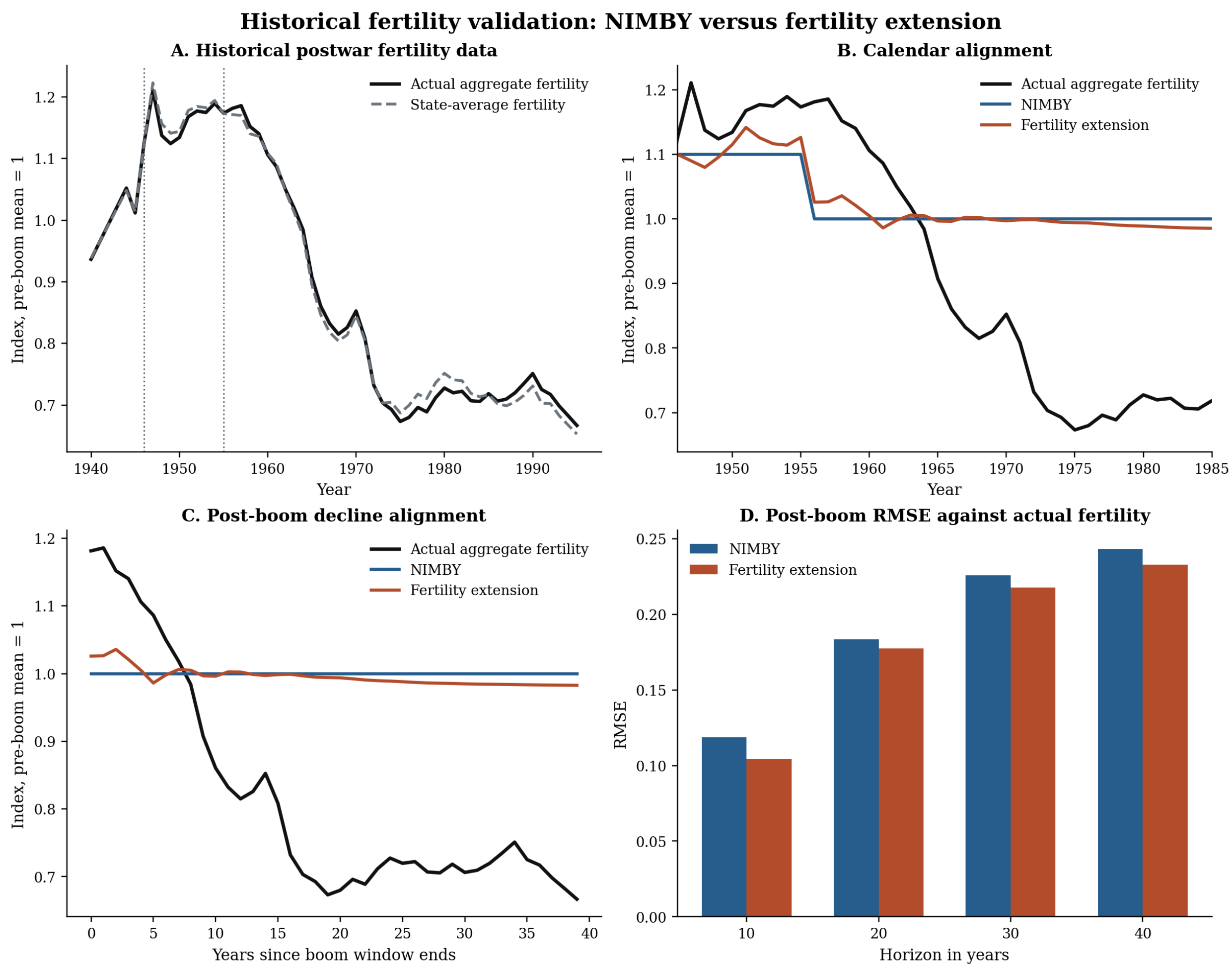

Figure 6: Historical fertility validation. The fertility extension fits the aligned post-boom decline modestly better than the flat return path of the benchmark model.

6 Annual Transition Extension

The five-year benchmark remains the paper's main steady-state object, but it compresses the early family-formation window into a coarse transition block. The annualised branch is useful because it opens that window back up. It now does so with a cleaner closure than before. On the upgraded annual permits-starts-completions-stock block, the paper now has a stationary annual no-drift endpoint plus a full rational-expectations transition around that endpoint. The annual branch is therefore no longer only a drifting short-horizon stress test. It is a stationary annual closure with additional short-window support experiments layered on top.

The annual transition still starts from a time-zero cross section that matches the observed age distribution of four balance-sheet states by construction: renter without debt, renter with debt, owner with mortgage, and owner outright. The annual cases are then frozen around two support regimes. The *benchmark* case uses qualification weight 0.25 and family-help weight 0.15. The *robustness* case uses qualification weight 0.40 and family-help weight 0.30. Table 8 collects the resulting annual transition calibration layer. Those targets still discipline the first five annual periods most tightly, but they no longer exhaust the annual branch because the same calibrated construction-flow block now admits a stationary annual endpoint.

Two reduced-form annual entry mechanisms are allowed to operate for young first-time buyers. The first is a broad help-to-buy / qualification margin that proxies for low-down-payment products, flexible underwriting, and co-borrower style qualification support. The second is a narrower family-help margin that acts like a deposit gift at the point of purchase. In the short-window annual exercises these remain transition objects: they explain how some young households still enter ownership with mortgage debt despite high house prices, not why house prices are high in the first place. In the paper's interpretation, high prices still come from persistent supply restriction; the annual entry channels are coping mechanisms.

The stationary no-drift endpoints on the calibrated construction-flow block are now economically sensible and numerically stable. In the benchmark case the annual stationary equilibrium has $q^{ss} \approx 1.974$ and young mortgaged ownership for ages 25–34 of about 0.269; the corresponding baseline and robustness endpoints are $q^{ss} \approx 1.993$ and 1.961. That stationary endpoint matters because it turns the annual branch into a genuine closure rather than a transition that only lives off an imposed drift path and an eventual price cap.

Figures 7 and 8 still report the drifted support experiments, but they should now be read as short-window policy diagnostics layered on top of the stationary annual closure rather than as the closure itself. Under the central +5 percent drift calibration, the benchmark regime raises the young owner share from 0.610 to 0.657 at $t = 5$ and raises the young mortgaged-owner share from 0.367 to 0.407. The robustness case is stronger, moving those same objects to 0.684 and 0.430. At $t = 20$ the effect is smaller but still positive. Across drift rates of +2, +5, and +8 percent, the benchmark and robustness regimes remain ordered the same way. So the support margins still matter most in the annual crunch window, even though the live annual closure is now the stationary no-drift branch.

The core annual RE object is now the stationary full-path transition from the observed snapshot to that endpoint. In the benchmark case the no-drift full-RE path gives $q_1 \approx 1.001$, $q_5 \approx 1.010$, $q_{10} \approx 1.031$, $q_{20} \approx 1.101$, and $q_{80} \approx 1.721$. The early and middle years are almost identical to the corresponding no-RE no-drift anchor, while by $t = 80$ the ownership distribution is essentially at its stationary endpoint and prices are still converging upward toward it. The annualised model can therefore now be described as a genuine full-RE annual transition around a stationary annual equilibrium.

For paper-facing exposition, the annual RE objects are best grouped into four layers. First, the *core closure* is the stationary annual endpoint plus the no-drift full-RE transition around it. That is the live annual equilibrium object. Second, the *simple bridge objects* ask what happens when one extra forward-looking margin is added without making the whole annual system recursive. The fertility-side bridge is the exogenous fertility-demand shock: relative to no-shock full RE, the benchmark price path is about 0.021 higher at $t = 5$ and about 0.028 higher at $t = 20$, with only tiny

Table 8: Annual transition calibration targets and frozen cases

Group	Moment	Target or role	Drift baseline	Benchmark	Robu
Snapshot matched by construction	Owner share 25-34 at t=0	0.407 observed t=0 snapshot	0.407	0.407	0.
	Mortgaged-owner share 25-34 at t=0	0.306 observed t=0 snapshot	0.306	0.306	0.
	Owner share 35-44 at t=0	0.605 observed t=0 snapshot	0.605	0.605	0.
	Mortgaged-owner share 35-44 at t=0	0.523 observed t=0 snapshot	0.523	0.523	0.
External incidence targets	Help-to-buy incidence over first 5 years	0.30-0.40 external first-time buyer help-to-buy band	–	0.181	0.
	Family-help incidence over first 5 years	0.20-0.30 external first-time buyer family-help band	–	0.105	0.
Main annual calibration window	Owner share 25-34 at t=5	short-run annual transition discipline	0.610	0.657	0.
	Mortgaged-owner share 25-34 at t=5	short-run annual transition discipline	0.367	0.407	0.
	Owner share 35-44 at t=5	short-run annual transition discipline	0.739	0.770	0.
	Mortgaged-owner share 35-44 at t=5	short-run annual transition discipline	0.315	0.338	0.
Persistence check only	Owner share 25-34 at t=20	persistence check, not main fitting target	0.680	0.718	0.
	Mortgaged-owner share 25-34 at t=20	persistence check, not main fitting target	0.320	0.347	0.
	Owner share 35-44 at t=20	persistence check, not main fitting target	0.841	0.867	0.
	Mortgaged-owner share 35-44 at t=20	persistence check, not main fitting target	0.196	0.205	0.

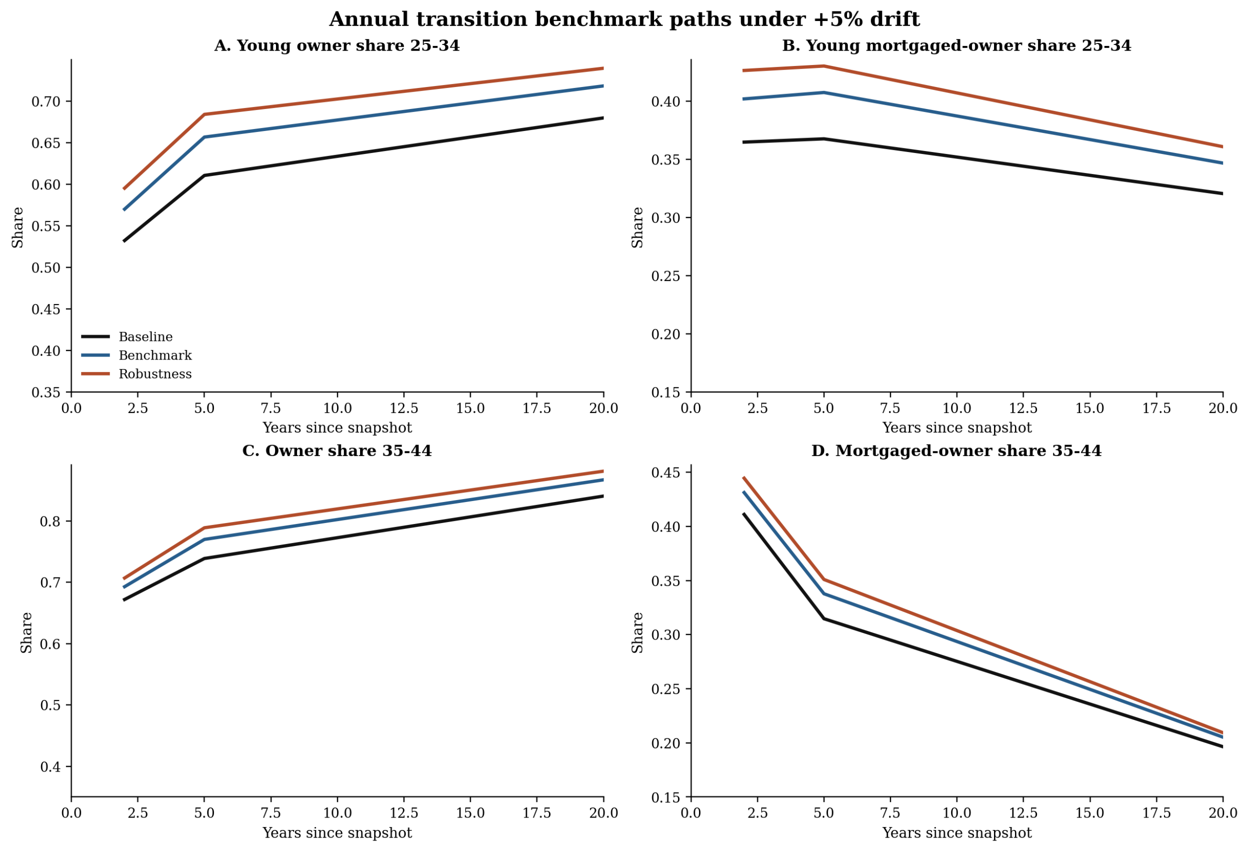

Figure 7: Annual transition benchmark paths under the central +5 percent drift calibration. These drifted support experiments are now read as short-horizon policy diagnostics layered on top of the stationary annual closure. The benchmark support regime raises young mortgaged ownership materially in the main calibration window, while the robustness case provides a stronger data-facing upper bound.

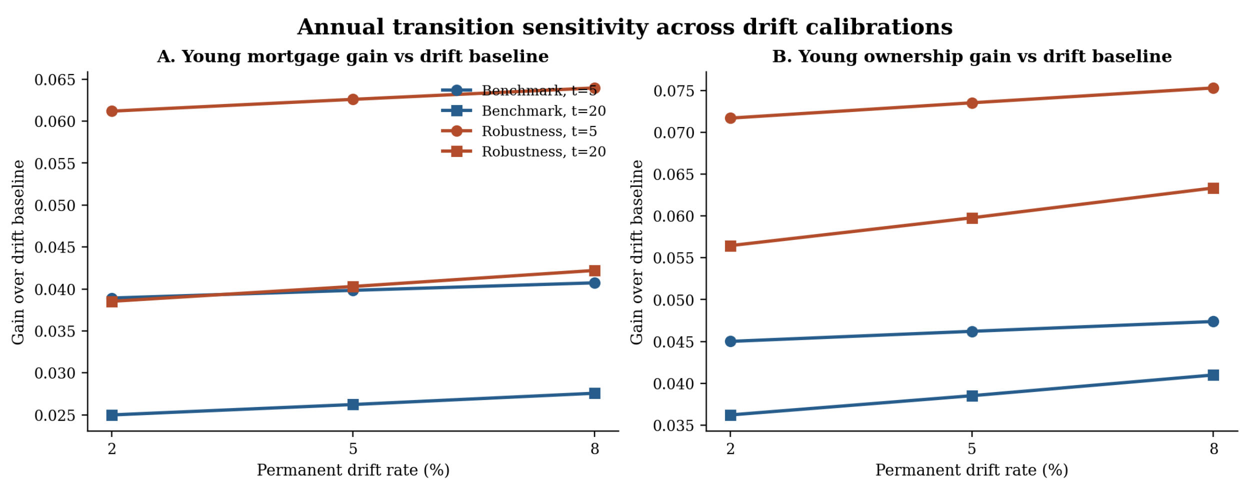

Figure 8: Annual transition sensitivity across drift calibrations. The benchmark and robustness regimes remain ordered the same way under +2, +5, and +8 percent annual housing drift. This bounded grid is now a robustness layer around the stationary annual full-RE closure.

changes in ownership composition. The policy-side bridge is the imposed regime-path RE exercise. A known switch from baseline support to benchmark support at $t = 5$ leaves the price path almost unchanged relative to the always-benchmark full-RE case, but it lowers young mortgaged ownership by about 0.026 at $t = 1$ and about 0.038 at $t = 5$ before the switch arrives.

Third, the *reduced-form recursive extensions* put forecasted aggregate states inside the fixed point without yet embedding the literal annual household recursion. The main "beyond-q-only" fertility object is the reduced-form endogenous fertility state. In the benchmark run the fertility multiplier reaches about 1.026 at $t = 5$ and 1.037 at $t = 20$, which pushes prices about 0.009 above the no-shock path at $t = 5$ and about 0.055 above it at $t = 20$, again with only modest ownership effects. On the policy side, the forecasted political-easing state, the recursive support-state object, and the joint fertility-plus- politics run all belong in this class: they are genuinely recursive in aggregate states, but those states are still reduced-form forecast rules rather than full household or voting problems solved anew each period.

Fourth, there is now a *structural-implied extension*. The structural-fertility operator version no longer feeds back only through an aggregate birth multiplier. It now also uses first-birth timing moments to tilt purchase transitions away from ages 25–34 and toward ages 35–44 as prices rise, while higher implied childlessness tilts entrant tenure composition toward renting. Quantitatively, the enriched structural operator lowers the benchmark annual RE price path by about 0.026 at $t = 20$, 0.113 at $t = 40$, and 0.349 at $t = 80$ relative to the no-structural RE benchmark. By $t = 80$ the benchmark young purchase scale is down to about 0.76, the mid-age purchase scale is up to about 1.15, and the entrant owner scale is down to about 0.82. That is a more genuinely micro structural object than the earlier birth-multiplier-only bridge. But it is still an operator bridge rather than a literal household-state recursion, because the Python annual branch does not yet carry explicit parity or children-at-home states.

The older drifting bounded-RE exercises therefore move down one level in the paper's hierarchy. The drifting $k = 3$ solve remains the main robustness case on the older annual pressure experiment; the drifting $k = 5$ case remains the aggressive robustness version; and the drifting $k = 20$ plus $T = 40$ full-path solves are appendix / feasibility material only. They still show that the upgraded stock-flow block cures the thin-price instability, but the stationary endpoint plus the $T = 80$ full-RE no-drift transition is now the live annual closure.

The annual branch therefore changes the interpretation of the model rather than overturning the five-year benchmark. The five-year branch remains the paper's main steady-state equilibrium object. The annual branch now adds a stationary annual closure, a model-consistent RE transition around that closure, and a short-horizon housing-access layer that isolates the crunch years the five-year model smooths over: early family formation, low liquid buffers, and young ownership entry with mortgage debt. In that sense, the annualised model extends the paper's main mechanism rather than replacing the baseline five-year result.

7 Conclusion

This paper asks whether housing scarcity can delay family formation and lower realized fertility. We develop a political-economy housing model in which endogenous land-use restrictions, set by majority vote, persistently constrain supply. The household problem includes endogenous births, children at home, and a crowding margin through which high house prices raise the effective cost of having a child.

Two results stand out. First, high house prices delay first births in the steady state. At the two benchmark crossing prices, moving from the fertility-model crossing to the higher NIMBY benchmark price delays mean first birth by about 1.4 years and raises the childless share from about 15 to 35 percent. Second, persistent housing scarcity lowers fertility robustly across three supply-tightening margins, each calibrated to a common house-price target. The effect is equivalent to roughly 0.05 TFR points—about 13 percent of the US fertility decline since 2007—and it compounds over time as demographic aging and restricted supply reinforce each other.

These results suggest a channel through which housing policy affects demographic outcomes. Restrictions on new construction do not merely raise house prices. By making space scarce, they delay and discourage family formation—particularly among younger and lower-income households who face the tightest housing constraints. As populations age and housing markets remain constrained, this channel is likely to grow in importance.

Our estimates are conservative. Several modeling choices each omit a channel that would amplify, not reduce, the housing–fertility effect; collectively they make the magnitudes we report a lower bound on the true effect of housing scarcity on family formation.

First, the price-sensitive birth cost $\kappa_q q_t$ is a reduced form for the resource cost of space reallocation at birth—moving to a larger unit, furnishing a child's room, or paying higher rent on the marginal square meter. A more structural treatment that embeds the reallocation decision explicitly in the housing adjustment problem would likely generate larger price sensitivity. Second, first-birth timing is disciplined by age-specific realized-birth shifters calibrated to the recent U.S. first-birth age distribution, but later-parity births use unit realized weights. Incorporating biological fecundity decline would amplify the cost of delay and widen the fertility gap between supply regimes. Third, the model treats the household as a single decision-maker. In practice, if the more housing-cost-sensitive partner holds effective veto power over birth decisions, the housing channel would be amplified (Doepke and Kindermann, 2019). Fourth, children do not affect household income. In reality, children reduce maternal labour supply, tightening the budget constraint and reinforcing the housing channel. Fifth, the leave-home schedule is exogenous. High housing costs cause adult children to stay home longer, further crowding the parental dwelling—Appendix D shows that shifting 20 percent of age-20 departures to age 25 increases the crowding penalty meaningfully—and this channel is omitted from the baseline.[2]

Each of these omissions works in the same direction. Including any one of them would strengthen the fertility–housing channel; including several simultaneously would likely produce effects meaningfully larger than those we report.

The annualised transition extension is no longer only a bounded short-horizon exercise. On the upgraded permits-starts-completions-stock block, the annual branch now has a stationary no-drift endpoint and a full $T = 80$ rational-expectations transition around that endpoint. Starting from the observed age-by-tenure snapshot, the support wedges are still most informative over a short 0–5 year crunch window: broad help-to-buy / qualification support does most of the work in sustaining young mortgaged ownership under pressure, while lighter family deposit help acts as an amplifier. Those annual support results are therefore best read as short-window policy diagnostics layered on top of a stationary annual closure, not as the source of a separate long-run annual benchmark.

The upgraded annual price block also changes the reading of the expectations problem. On the old thin annual bridge, richer rational-expectations layers quickly produced implausible price movements. Once politics feed permits, permits feed completions, completions feed stock, and prices clear against stock, the annual branch admits not only tame bounded RE but also a stationary full-path RE closure. The older drifting $k = 3$ and $k = 5$ exercises now read as robustness cases, while

[2] The model uses five-year periods, which is coarse for birth timing. The logistic birth choice generates continuous variation in birth probabilities across age groups, so the 1.4-year mean delay between equilibria reflects a fractional shift in the age-specific distribution rather than a discrete five-year jump. Annual periods would multiply the state space fivefold; the five-year discretisation is standard in life-cycle housing models.

the endogenous-fertility annual RE object mainly shifts prices rather than ownership composition. So the earlier annual RE pathology appears to have been mainly a thin price-block problem rather than an unavoidable consequence of annual timing itself.

A Relation to the benchmark model

The fertility model nests the benchmark of Gross and Chivers (2025) exactly. When the fertility block is shut off, the fertility solver reproduces the upstream benchmark rather than merely approximating it. This matters because it ensures the comparison is not driven by a different political closure, a different real-estate block, or a different equilibrium definition.

Table 9 reports the benchmark crossing comparison. The political crossing shifts from 2.326 in the benchmark model to 1.750 once fertility is active under the refreshed timing benchmark. The normalized vote-per-unit-mass object is the preferred political-support measure; raw vote totals are retained only for continuity with the upstream benchmark. On the current benchmark grids, this convention does not change the crossing prices because stationary mass is constant within each model. Figure 9 shows the corresponding steady-state objects on the common house-price grid.

Table 9: Steady-state benchmark comparison

Object	Benchmark	Fertility extension
Crossing house price	2.326	1.750
Vote at refined crossing price	0.228	−0.004
Vote per unit mass at refined price	0.036	−0.004
Stationary household mass	6.388	1.000
Average birth rate	–	0.523

B Wide-range comparative statics

Table 10 reports the timing and parity gradients across a wider range of house prices than the equilibrium comparison in the main text. These comparative statics illustrate the mechanism's shape but do not correspond to any particular equilibrium.

Table 10: Steady-state first-birth timing and parity distribution (wide range)

	First-birth timing			Parity distribution at age 50			
House price q	Mean age	Share 30+	First-birth rate	Childless	1 child	2 children	3+ children
1.50	25.83	0.139	0.226	0.095	0.016	0.344	0.546
1.75	26.22	0.201	0.213	0.148	0.027	0.407	0.417
2.00	26.74	0.281	0.194	0.223	0.045	0.442	0.290
2.50	28.13	0.477	0.145	0.419	0.092	0.388	0.101
3.00	29.79	0.671	0.098	0.607	0.133	0.236	0.024

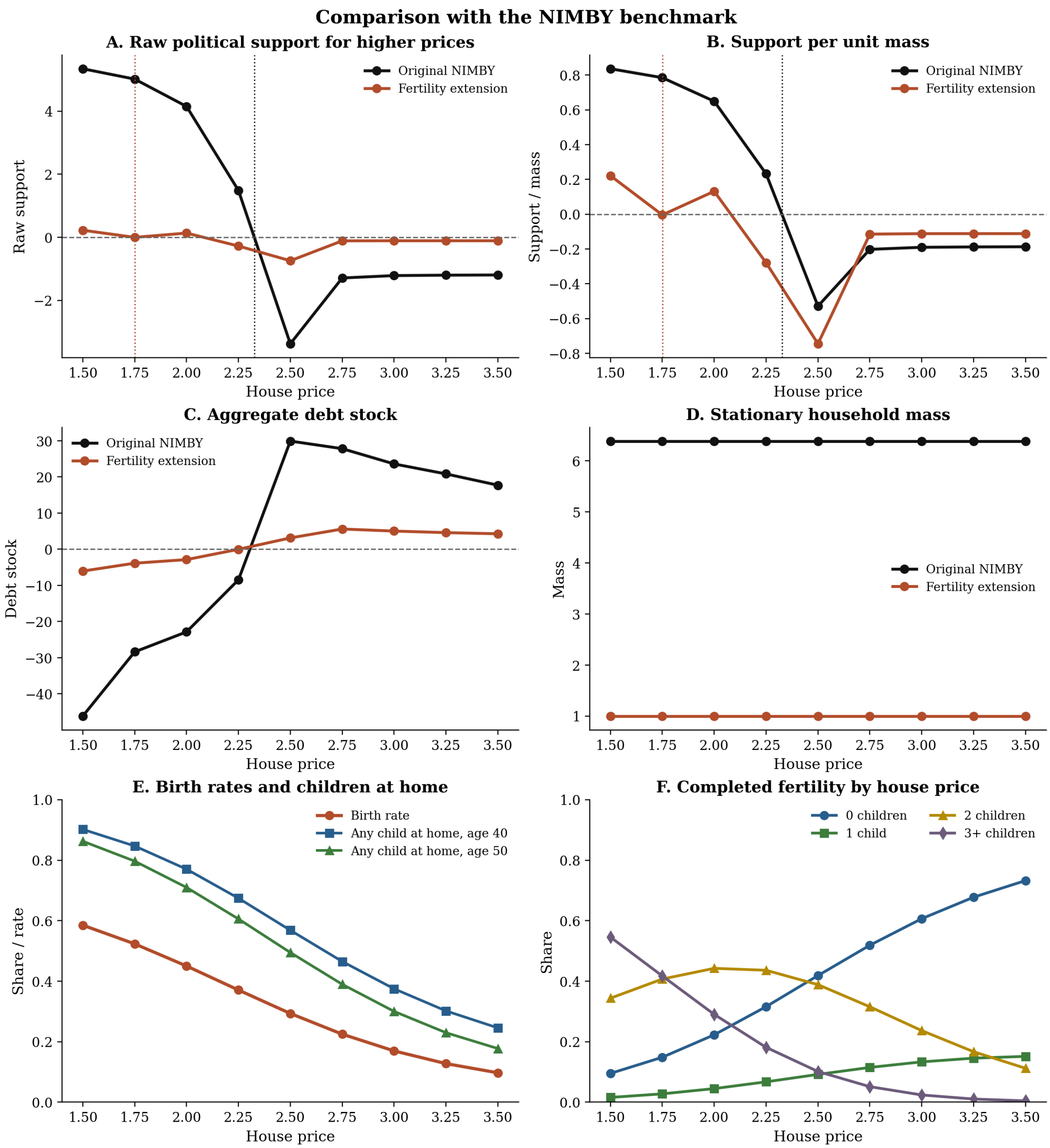

Figure 9: Steady-state benchmark comparison on the common house-price grid. The fertility extension shifts the aggregate support schedule toward lower prices and changes family outcomes along the grid. The normalized vote-per-unit-mass panel is the preferred cross-model political-support comparison.

C Mechanism parameter sensitivity

Table 11 reports the sensitivity of the main fertility outcomes to one-at-a-time perturbations of the four mechanism parameters: the price-sensitive birth cost κ_q, the crowding parameter λ_c, the logistic scale σ, and the first-birth utility $\phi(0)$. Each row perturbs one parameter from its benchmark value (shown in bold) while holding the others fixed and re-solves for the political crossing price.

Table 11: Mechanism parameter sensitivity

Parameter	Value	Crossing q	Mean first-birth age	Childless share	First-birth rate	Avg birth rate
Price-sensitive birth cost κ_q						
	0.18	—	—	—	—	—
	0.24	1.750	26.22	0.148	0.213	0.523
	0.30	—	—	—	—	—
Crowding parameter λ_c						
	0.12	—	—	—	—	—
	0.18	1.750	26.22	0.148	0.213	0.523
	0.24	—	—	—	—	—
Logistic scale σ						
	0.10	—	—	—	—	—
	0.15	1.750	26.22	0.148	0.213	0.523
	0.20	—	—	—	—	—
First-birth utility $\phi(0)$						
	0.90	—	—	—	—	—
	1.05	1.750	26.22	0.148	0.213	0.523
	1.20	—	—	—	—	—

These comparative statics do not correspond to any particular calibration target; they show how the mechanism responds to variation in its key primitives. The sensitivity table will be completed once the computational sweep (Appendix C) is finished.

D Endogenous departure envelope

The benchmark model treats the departure schedule as exogenous. In practice, high house prices may cause adult children to stay home longer, amplifying the crowding penalty. This appendix provides a back-of-the-envelope calculation of the quantitative significance of this channel.

The benchmark departure schedule assigns probabilities $[0.588, 0.147, 0.265]$ at child ages $[20, 25, 30]$. Suppose high house prices shift 20 percent of age-20 departures to age 25, giving a delayed schedule of $[0.470, 0.265, 0.265]$. At the benchmark crossing price ($q = 1.750$), the expected number of children at home at parent age 40 is approximately 1.047 under the benchmark schedule. Under the delayed schedule, more children remain at home at each parent age because departures are shifted later.

The change in the crowding penalty is

$$\frac{(1 + \lambda_c\, h_{\text{delayed}})^{\psi_c}}{(1 + \lambda_c\, h_{\text{benchmark}})^{\psi_c}} - 1,$$

where $\lambda_c = 0.18$ and $\psi_c = 0.70$. This calculation will be completed once the exact children-at-home profiles are extracted from the sensitivity script. The purpose is to bound the quantitative importance of endogenous departure, not to re-solve the model with a fully endogenous departure schedule.

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